Small Signal Modelling of PSFB-II

Introduction

In the realm of power electronics, ensuring stable and precise output voltage from converters is paramount, especially in critical applications like charging infrastructure. This lecture delves into the intricate process of small signal modeling for the Phase Shifted Full Bridge (PSFB) converter, a crucial step towards designing effective closed-loop control systems. While the PSFB converter offers advantages like Zero Voltage Switching (ZVS), its operational characteristics, particularly the presence of a "D-loss" term, introduce complexities that necessitate a detailed small signal analysis. Understanding how small perturbations in duty cycle, input voltage, or load current affect the output allows engineers to predict system behavior, design robust controllers, and ultimately guarantee the desired performance and reliability of the power supply. Our primary objective is to derive the transfer function, G_VD(s), which relates changes in output voltage to changes in the duty ratio, forming the bedrock for closed-loop regulation.

Core Concepts

Term	Definition	Significance
`D-loss`	A duty cycle loss term in PSFB converters, primarily due to leakage inductance and switching frequency, which reduces the effective duty cycle.	It directly impacts the converter's effective duty cycle and output voltage, making the control more complex than ideal converters.
`D_effective`	The actual duty cycle experienced by the converter, calculated as the nominal duty cycle minus the `D-loss` term.	This is the true control variable influencing the output, and its dependence on operating conditions must be accounted for in modeling.
`Small Signal Approximation`	A linearization technique where small perturbations around an operating point are assumed, allowing non-linear	Simplifies complex non-linear converter equations into linear models, enabling the use of linear control

Term	Definition	Significance
	terms involving products of small signals to be neglected.	theory for analysis and design.
`Steady-State`	The operating condition where all system variables (voltages, currents) have reached their average DC values and are no longer changing over time.	Provides the DC operating point around which small signal perturbations are applied, crucial for defining the linearization point.
`Average Large Signal Model`	A set of differential equations describing the average behavior of converter voltages and currents over a switching period, neglecting high-frequency ripple.	Forms the basis for small signal analysis by providing the fundamental dynamic relationships before perturbation and linearization.
`Buck-Derived Topology`	A classification for converters whose output stage operation, particularly the inductor and capacitor, mimics that of a simple buck converter.	Simplifies the small signal modeling process by allowing the use of established buck converter models as a starting point.
`G_VD(s)`	The transfer function relating the small signal change in output voltage (V_out(s)) to the small signal change in duty ratio (D(s)).	This is the primary goal of small signal modeling, essential for designing feedback controllers to regulate the output voltage.
`DI and DV terms`	Additional terms in the `D_effective` equation that capture the change in `D-loss` due to changes in inductor current (DI) and input voltage (DV).	These terms highlight the inherent coupling and non-linearity in PSFB converters, requiring careful incorporation into the small signal model.

Detailed Analysis

1. The Impact of D-loss on Effective Duty Cycle

The PSFB converter's operation is significantly influenced by a phenomenon known as "D-loss," which effectively reduces the duty cycle applied to the output stage. This `D-loss` is primarily attributed to the leakage inductance of the transformer and the switching frequency, and it is inversely proportional to the input voltage (`Vin`) and directly proportional to the inductor current (`IL`). Consequently, the actual effective duty cycle (`D_effective`) seen by the output is `D - D_loss` .

- **Dependence of D-loss**: The `D-loss` term is not constant; it varies with both the inductor current (`IL`) and the input voltage (`Vin`).
- A small change in `IL` leads to a small change in `D-loss` , which in turn alters `D_effective` . This effect is captured by the `DI` term.
- Similarly, a small change in `Vin` causes a small change in `D-loss` , impacting `D_effective` . This is represented by the `DV` term.
- **Implications for Regulation**: When the load changes, the inductor current changes. To maintain a regulated output voltage, the duty ratio must adjust. This adjustment, however, affects the voltage across the inductor, which then changes the inductor current, further altering the `D-loss` and `D_effective` . This complex interplay necessitates a small signal model that can accurately capture these dynamic relationships.

2. Objective: Deriving the Output Voltage to Duty Cycle Transfer Function

The ultimate goal of this small signal modeling exercise is to determine the transfer function `G_VD(s) = V_out(s) / D(s)` . This transfer function describes how small variations in the duty ratio (`D`) translate into small variations in the output voltage (`V_out`) in the s-domain. This is a critical component for designing a closed-loop control system that can regulate the output voltage to a desired value despite disturbances. The duty ratio in a PSFB converter is controlled by adjusting the phase shift between the diagonal switches (e.g., S1 and S4), which effectively modulates the voltage applied to the output rectifier.

3. Leveraging the Buck Converter Analogy

To simplify the modeling process, the PSFB converter's output stage is recognized as a "buck-derived topology." This means that the behavior of the output inductor and capacitor, which filter the rectified voltage, is analogous to that of a standard buck converter.
* **Operational Similarity**: During the `D*Ts/2` period, the PSFB applies a voltage

proportional to `N*Vin` across the output inductor, similar to how a buck converter applies `Vin`. During the `(1-D)*Ts/2` period, the diode conducts, applying `-V_out` across the inductor, identical to a buck converter.
* **Modeling Approach**: By first deriving the small signal model for a generic buck converter with an input voltage `N*Vin` and then incorporating the PSFB-specific `D_effective` and its dependencies (`DI` , `DV` terms), we can efficiently arrive at the PSFB's small signal model.

4. Small Signal Modeling Methodology

The general approach to small signal modeling involves several key steps:
1. **Average Large Signal Model**: First, develop a set of equations that describe the average behavior of the converter's voltages and currents over a switching period. This model ignores high-frequency ripple but captures the fundamental dynamics.
2. **Small Perturbation**: Introduce small, time-varying perturbations to the steady-state DC operating point of each variable (e.g., `D = D + d_tilda` , `V_C = V_C + v_c_tilda`).
3. **Linearization**: Apply the small signal approximation, where products of two small signal quantities are considered negligible (e.g., `d_tilda * i_l_tilda ≈ 0`). This linearizes the non-linear average model around the operating point.
4. **Equivalent Circuit/Transfer Function**: Convert the linearized equations into an equivalent small signal circuit or derive the transfer function in the s-domain.

5. Deriving the Average Large Signal Model for the Buck Converter

Considering a buck converter with an input voltage `N*Vin` (representing the rectified voltage in PSFB), the dynamic equations for the inductor current (`IL`) and capacitor voltage (`VC`) are derived:
* **During `D*Ts` period (switch S is ON)**:
* `L * dIL/dt = N*Vin - V_out`
* `C * dVC/dt = IL - V_out/RL`
* **During `(1-D)*Ts` period (diode D is ON)**:
* `L * dIL/dt = -V_out`
* `C * dVC/dt = IL - V_out/RL`
* **Averaging the Equations**: By averaging these over a switching period `Ts` , we obtain the average large signal model:
* `L * dIL/dt = N*Vin*D - V_out`
* `C * dVC/dt = IL - V_out/RL`
* Additionally, `V_out = VC` and `I_in = D * IL` .

6. Linearization using Small Perturbation

To linearize the average large signal model, each variable is expressed as a sum of its steady-state (DC) component and a small, time-varying (AC) perturbation:
* `D = D + d_tilda`
* `VC = VC + v_c_tilda` (where `VC` is the steady-state capacitor voltage)
* `IL = IL + i_l_tilda` (where `IL` is the steady-state inductor current)
* `N*Vin = N*Vin + n_v_in_tilda` (where `N*Vin` is the steady-state input voltage)

The small signal approximation states that products of two small signal quantities are negligible (e.g., `d_tilda * i_l_tilda ≈ 0`).

7. Deriving Small Signal Equations

Substituting the perturbed variables into the average large signal model and applying the small signal approximation, we separate the steady-state and small-signal components. The steady-state equations are obtained by setting all `d/dt` terms and small signal terms to zero:
* `0 = N*Vin*D - V` (where `V` is steady-state output voltage)
* `0 = IL - V/RL`
* `V = VC`
* `I_in = D * IL`

The small signal equations, after canceling out the steady-state terms and neglecting products of small signals, become:
* `L * di_l_tilda/dt = -v_c_tilda + N*Vin*d_tilda + n_v_in_tilda*D`
* `C * dv_c_tilda/dt = i_l_tilda - v_c_tilda/RL`
* `v_c_tilda = v_out_tilda`
* `i_in_tilda = d_tilda*IL + D*i_l_tilda`

8. Equivalent Small Signal Circuit for the Buck Converter

From these small signal equations, an equivalent circuit can be constructed. This circuit represents the dynamic behavior of the converter for small perturbations.
* The first equation (`L * di_l_tilda/dt`) represents the voltage across the inductor, which is `(-v_c_tilda + N*Vin*d_tilda + n_v_in_tilda*D)`. This implies a series connection of voltage sources and the inductor.
* The second equation (`C * dv_c_tilda/dt`) represents the current through the capacitor, which is `(i_l_tilda - v_c_tilda/RL)`. This implies a parallel connection of the capacitor, a current source (`i_l_tilda`), and the load resistor (`RL`).
* The `i_in_tilda` equation suggests a current source `d_tilda*IL` and a current `D*i_l_tilda` on the input side, often represented by an ideal transformer with a turns ratio of `1:D` to couple the input and output sides.

The basic buck converter small signal model includes:
* A voltage source `N*Vin*d_tilda` (duty cycle modulation)
* A voltage source `D*n_v_in_tilda` (input voltage modulation)
* The inductor `L`
* The capacitor `C`
* The load resistor `RL`
* A current source `d_tilda*IL` (duty cycle modulation of input current)

9. Incorporating PSFB Specifics: The DI and DV Terms

The buck converter model is then adapted for the PSFB by incorporating the `DI` and `DV` terms, which account for the change in `D_effective` due to changes in inductor current and input voltage.
* The `D_effective` in PSFB is `D - D_loss`. When perturbed, `d_effective_tilda = d_tilda - d_loss_tilda`.
* The `d_loss_tilda` term is a function of `i_l_tilda` and `n_v_in_tilda`. Specifically, `d_loss_tilda = DI * i_l_tilda + DV * n_v_in_tilda`.
* Therefore, the effective duty cycle perturbation `d_effective_tilda` becomes `d_tilda - (DI * i_l_tilda + DV * n_v_in_tilda)`.
* This modified `d_effective_tilda` replaces `d_tilda` in the buck converter's small signal equations, leading to additional controlled voltage and current sources in the equivalent circuit.
* The final PSFB small signal equivalent circuit will feature:
* A voltage source `N*Vin*d_tilda`
* A controlled voltage source `-(DI + DV)*N*Vin` in series with the inductor, representing the impact of `D-loss` on the effective voltage.
* A controlled current source `(DI + DV)*IL` on the input side, representing the impact of `D-loss` on the input current.
* The inductor `L`, capacitor `C`, and load `RL`.
This comprehensive equivalent circuit for the PSFB converter's small signal model is then used to derive the desired transfer functions for control design.

Key Takeaways

- [ ] The PSFB converter's effective duty cycle (`D_effective`) is reduced by a `D-loss` term, which depends on inductor current and input voltage.
- [ ] Small signal modeling linearizes the converter's behavior around an operating point, enabling the use of linear control theory.
- [ ] The PSFB converter's output stage can be modeled as a buck-derived topology, simplifying the initial derivation.

- [ ] The modeling process involves deriving an average large signal model, applying small perturbations, and linearizing the equations.
- [ ] The small signal approximation allows neglecting products of two small signal quantities (e.g., `d_tilda * i_l_tilda ≈ 0`).
- [ ] The `DI` and `DV` terms, representing the change in `D-loss` due to changes in inductor current and input voltage, are crucial for an accurate PSFB model.
- [ ] The final small signal model for PSFB incorporates controlled voltage and current sources to account for the `D-loss` effects.
- [ ] The primary goal is to obtain the `G_VD(s)` transfer function, which is essential for designing closed-loop controllers.

Conclusion

The detailed small signal modeling of the PSFB converter, culminating in its equivalent circuit, is a foundational step in designing high-performance power supplies for charging infrastructure. By meticulously accounting for the `D-loss` phenomenon and its dependencies on inductor current and input voltage, we move beyond idealized models to capture the true dynamic behavior of the converter. This linearized model, represented by the equivalent circuit, provides the necessary framework to derive critical transfer functions like `G_VD(s)`. With these transfer functions in hand, engineers can proceed to design robust closed-loop control systems, ensuring that the output voltage remains stable and regulated at the desired value, even under varying load conditions or input voltage fluctuations. The next logical step involves using this model to analyze stability, design compensators, and ultimately implement a reliable control strategy for the PSFB converter.